

ASIF PERVEZ POLOK

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🐙 [Github](#)

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EDUCATION

Noakhali Science and Technology University

Dec. 2017 – Feb. 2023

Bachelor of Science in Information and Communication Engineering (ICE)

CGPA - 3.26 out of 4.00

Courses - Artificial Intelligence, Digital Image Processing, Calculus and Differential Equation, Statistics for Communication Engineering, Geometry and Vector Analysis, , Information Theory

EXPERIENCE

Machine Learning Engineer

Dec 2023 – Present

Business Automation Ltd

Rajshahi, Bangladesh

- Developing cutting-edge solutions in Generative AI, Computer Vision, and Deep Learning, leveraging expertise in Neural Networks, image processing, and machine learning to drive AI innovation. Contributing to the development of GenAI models on custom data while ensuring seamless deployment and scalability.
- Advanced AI Technologies – Proficient in Graph Knowledge & Retrieval-Augmented Generation (RAG) for enhanced AI reasoning, Agentic AI & Multi-Agent Systems for autonomous decision-making, Layout-based Segmentation & OCR, and LLM Fine-tuning for domain-specific applications.
- AI Deployment & MLOps – Hands-on experience in API creation, Dockerization, CI/CD pipelines, and scalable infrastructure for deploying production-ready AI solutions.

Artificial Intelligence Training

June 2023 – Sept 2023

Extent IT Ltd

Dhaka, Bangladesh

- 4-month professional "Artificial Intelligence" training program funded by EDGE, IIT, and BCC. Engaged as an AI Trainee focused on algorithm development in Machine and Deep Learning platform. Led data preprocessing, exploratory data analysis, and feature engineering to boost model accuracy and performance. Explored NLP and Computer vision core topics and Contributed to projects by applying advanced analytical techniques and statistical models, extracting actionable insights from complex datasets to drive impactful results.

Artificial Intelligence Intern (Remote)

Sept 2022 – Dec 2022

Shiga Corp & Trust

Tamil-Nadu, India

- Completed a 4-month remote internship funded by Shiga Corp & Trust Ltd., focusing on cutting-edge technologies in Machine Learning, Deep Learning, Time Series Analysis, and NLP. The internship involved specialized coursework that significantly enhanced technical proficiency and deepened expertise in advanced AI methodologies.
- Projects:
 - Fashion Recommendation System | *ResNet50, K Nearest Neighbor, Streamlit* | [🐙 GitHub](#)
 - Chatbot Building Deep Learning Project | *Bag of Words, ANN, Tkinter* | [🐙 GitHub](#)

Research Assistant (Remote)

Nov 2022 – Feb 2023

The University of Texas at El Paso

Texas, United States

- Collaborated with a Ph.D. student from The University of Texas at El Paso on academic research and assignments, contributing to the development of some research papers as well as the formulation of research hypotheses and methodologies.
- Research Projects:
 - Sartorious Cell Segmentation and Masking | *Tensorflow, UNet, CNN, Data Augmentation* | [🐙 GitHub](#)
 - Fine-tuning Bangla-Bert for Bangla News headline Classification | *Pytorch, BERT, NLTK* | [🐙 GitHub](#)

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

AI Stacks: Machine Learning, Deep Learning, NLP & Computer Vision

Generative AI: Prompt Engineering, LLMs, VLMs, RAG, Vector Database, Knowledge Graph (Neo4j), Langchain, OpenAI, Gemini, Groq, Ollama, OCR Models

Agentic AI: Langgraph, CrewAI, Phidata

MLOps: MLflow, Git & Github, CI/CD(Jenkins, Github Action), Docker, Kubernetes, Prometheus, Grafana, XAI, Whylab & Whylog

Data Analysis & Visualization: Tableau, Pandas, Numpy, Matplotlib, Seaborn, Plotly

Database: MySQL, PostgreSQL, MongoDB

Deployment: Flask, Streamlit, FastAPI

Others: Web Scraping(Scrapy), Time Series Analysis

PROJECTS

AIAgentInsight: Multi-Agent Product Analyzer | *Langchain, LLama3.3, Crawl4ai, Streamlit* | [🔗 GitHub](#)

- * AIAgentInsight is a Streamlit-based web application designed to provide detailed insights and recommendations for analyzing product or company-related queries. It utilizes various agents for scraping, validating, and summarizing data from platforms like G2, Crunchbase, and DuckDuckGo, combined with the power of LLMs for query analysis and summary generation.

Neo4j-Driven AI Chatbot with Knowledge Graph and LLM Integration | *LLM, Neo4j, HybridSearch* | [🔗 GitHub](#)

- * Developed a chatbot system that combines Neo4j Knowledge Graphs, Retrieval-Augmented Generation (RAG), and Hybrid Search to provide contextually accurate responses across PDF, CSV, and TXT datasets. Integrated graph-based reasoning and generative AI for dynamic, real-time querying and scalable conversational experiences, ideal for enterprise applications.

DocuFix: Intelligent Document Detection & Rotation Correction | *YoloV11, Flask, OpenCV* | [🔗 GitHub](#) | [🌐 Live](#)

- * Developed an intelligent document segmentation and rotation correction solution, utilizing the YOLOv11 segmentation model for document extraction and OpenCV for image processing tasks. Implemented automatic rotation correction to ensure proper alignment, integrated the solution into a Flask web application, and deployed it using Docker.

WhisprNote: Minutes of Meetings from Audio and Video Recording | *Whisper, Gemini-pro* | [🔗 GitHub](#)

- * Designed an audio-visual transcription system leveraging Whisper and LLM models to generate clear, actionable minutes of meetings and summarizes meeting transcript. Ensured alignment and accountability by capturing key decisions and action plans in an engaging, accessible format.

Name Entity Recognition (NER) with MISTRAL, BERT, and FLAN T5 | *LLMs, Unsloth* | [🔗 GitHub](#)

- * Developed Named Entity Recognition (NER) system using MISTRAL, BERT, and FLAN T5 models. The Bangla-Bert model was used to identify and classify entities like person, organizations, locations, objects, and emails, while MISTRAL and FLAN T5 were fine-tuned on a Cyber Security dataset to classify software and security-related entities

Automated Visiting & NID Card Cropping System | *YoloV8, OpenCV, Flask* | [🔗 GitHub](#) | [🌐 Live](#)

- * Developed a web application using Flask for detecting and cropping visiting and NID cards from uploaded images, integrating OpenCV for image processing and employing a fine-tuned Ultralytics YOLOv8 model with real card and NID images to enhance detection accuracy and seamlessly crop detected regions.

Intelligent Question Answering and Code Generation Chatbot for Tabular Data | *OpenAI,LLM* | [🔗 GitHub](#)

- * Developed an intelligent system that enables users to ask natural language questions about tabular data, system able to generates code snippets to provide accurate answers. The system supports data formats like CSV and Excel that allowing users to easily work with and analyze multiple datasets with ease. System also creates visualizations based on queries to enhance the interpretation of data insights according to user inputs.

BanglaSpellT5: Spell Correction for Bangla Language | *Bangla-T5, HappyTransformer, NLTK* | [🔗 GitHub](#)

- * Developed a spell correction system using the Bangla-T5 model by executing comprehensive data preprocessing, model training, and evaluation, resulting in enhanced accuracy and significant improvements in Bangla spell correction tasks.

Image To Caption Generator DeepLearning Project | *Encoder, Decoder, Attention, ResNet101, LSTM* | [🔗 GitHub](#)

- * Utilized a pre-trained CNN (ResNet101) as an encoder to extract features from images for generating descriptive captions and an LSTM decoder to generate captions based on the extracted image features, forming a complete image captioning system using deep learning techniques.

CERTIFICATIONS

- Artificial Neural Networks (ANN) with Keras in Python and R | Udemy | [view](#)
- Python for Deep Learning: Build Neural Networks in Python | Udemy | [view](#)
- Time Series Analysis and Forecasting using Python | Udemy | [view](#)
- ICEPC Farewell Contest 2022 organiser award | ICE, NSTU | [view](#)

ACHIEVEMENTS & LEADERSHIP

- Kaggle Notebook Expert | Highest Rank: 1693 | Upvotes: 244 | [k Kaggle](#)
- ICE Sports Club, NSTU | **President** | **Jan 2022 – Feb 2023**

REFERENCE

• Md Easin Hasan, Ph.D.

Data Scientist (ORISE Fellow), Food and Drug Administration, USA

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The University of Texas at El Paso, United States